

Recovery

with SHONA HALSON and PHIL GLASGOW

Paul Quantrill will never forget the feeling of pure exhaustion at the most important moment of his career. It was the fourth game of the 2004 American League Championship Series, at Fenway Park, with the Yankees trying to finish a sweep of the Boston Red Sox. The teams were tied, 4-4, heading into the bottom of the 12th. It was Quantrill's 100th relief inning of the season. 'We were done,' Quantrill said. 'We were wrecked. We battled, but we just weren't where we needed to be.'

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The concept of mechanotransduction (Chapter 17) recognises that when the body is exposed to mechanical stress, it adapts. In sport, careful manipulation of training parameters induces specific adaptations to enhance performance. Training-induced adaptations include improvements in strength, endurance, neuromuscular control and flexibility. Training to promote adaptation requires a delicate balance of overload and recovery. Strategies to expedite recovery are central to training and competition plans.

Effective recovery can permit greater levels and quality of training and thereby enhance performance. Conversely, inadequate recovery from training and competition impair performance and may eventually result in injury, illness or burnout. How performance changes in response to exercise depends on a complex interplay of factors that include peripheral muscle changes,¹ central neural drive² and psychological responses.³ Effective recovery should seek to address the physiological, structural, neural and psychological limitations of performance.

A range of modalities have been purported to facilitate various aspects of recovery; the evidence for many of these modalities is scant and at times conflicting.

PRACTICE PEARL

As with many aspects of sports medicine, the most appropriate blend of recovery strategies depends on a detailed understanding of the sport and the athlete as well as the relative benefits of the choice of recovery modality.

ASSESSING RECOVERY

The traditional way to assess the effectiveness of interventions has been to test the athlete and investigate markers of the 'damage' associated with the test. The primary test models are (i) eccentric contraction induced delayed onset muscle soreness (DOMS); and (ii) sports-related training sessions. The variable nature of the exercise regimens used to induce damage and/or physiological stress make effective generalisation of findings challenging in some cases.

Following such a test, a typical test battery would include analysis of:

- general muscle function (peak torque maximum voluntary contractions: isometric and isokinetic)
- specific performance tasks (counter-movement jump (CMJ), squat jump, repeated sprint ability) which may be more meaningful in terms of direct impact on performance
- extent of tissue damage (creatine kinase, CK)
- level of metabolite removal (blood flow, blood lactate) and perception of soreness (pain on visual analogue scale, VAS)
- athlete's self-perception of recovery as well as psychological measures (e.g. profile of mood states, POMS).

ACTIVE RECOVERY

Active recovery refers to low-intensity work performed after exercise to facilitate better performance in a subsequent exercise bout. This is a common part of training; 81% of French professional soccer teams carry out some form of cool

down after training.⁴ Active recovery is therefore often the first form of recovery performed after exercise.

The type of active recovery performed will be heavily influenced by the duration, intensity and nature of the exercise from which the athlete is recovering. For example, the goals of recovery after a short maximal bout of exercise such as sprinting will be different to those following prolonged endurance events.

After high-intensity short-duration exercise

For recovery after short-duration high-intensity exercise, passive recovery appears to better promote resynthesis of phosphocreatine (PCr)—a rapidly mobilised reserve of high-energy phosphates in skeletal muscle—than does active recovery. Time to exhaustion was longer when intermittent exercise alternated with passive recovery (15s work: 15 s rest) than with active recovery.^{5, 6} Similar observations were reported for short cycle sprints (4s duration) when followed by passive recovery of 21 seconds when compared to low-intensity active recovery (32% VO_2max).⁷ Since recovery following repeated sprint activities involving the PCr system relies almost exclusively on aerobic factors,⁸ the better outcomes with passive recovery may be due to greater availability of oxygen for PCr resynthesis in resting muscles than when muscles are still being contracted (active).^{7, 9, 10} Passive rather than active recovery is therefore recommended following high-intensity short-duration exercise.

After longer-duration exercise—active recovery and metabolite clearance

Active recovery may have a greater role to play following longer duration activities. It may provide a benefit by increased removal of metabolites, particularly blood lactate. Muscle performance is impaired in the presence of high blood lactate levels,¹¹ so lactate level is widely considered to be important in determining the extent of recovery after high-intensity exercise. While the role of lactate in sports performance is still a subject of much debate, most studies consider it to be a useful indirect marker of cell and metabolic acidosis.¹²

Active recovery is the most effective way to increase blood flow and facilitate removal of metabolites following exercise.^{13–15} Exercise intensity levels close to the anaerobic threshold have been identified as most effective for lactate clearance.^{15–17} In a study of 10 moderately trained males, lactate levels were reduced most rapidly when participants ran at 80–100% of their lactate threshold.¹⁷ Runners who self-regulated their speed tended to run at around 80% of threshold and experience comparable reductions in blood lactate. This suggests that moderately trained individuals may self-select the

most appropriate exercise intensity. The physiological mechanism underpinning this observation is unclear.

Active recovery may be of greatest benefit in sports that involve a significant anaerobic contribution. It appears that active recovery is particularly effective if subsequent bouts of exercise are carried out within 10–20 minutes.^{15, 18} Given that blood lactate levels return to baseline within one hour of passive recovery (Fig. 13.1),^{16, 19} there is less convincing evidence for active recovery carried out several hours or days after exercise. In a series of studies investigating the effects of active recovery between games separated by 3 days in female soccer players, there was no influence on a wide range of measures of recovery including perceived muscle soreness, neuromuscular fatigue (CMJ), sprint performance, maximal isokinetic knee flexion and extension) or biochemical markers (inflammatory plasma cytokine response, plasma endogenous antioxidants, dietary antioxidants, oxidative stress markers, CK, urea).^{20, 21}

Similarly, in a study investigating self-selected recovery modalities in Australian Football League (AFL) players over a season, fewer players chose active recovery (pool or cycle) compared with cold-water immersion (CWI), compression garments or passive stretching.²² There were no benefits on performance measures during a season for any of the recovery modalities. However, those players who used CWI, compression and floor stretching reported

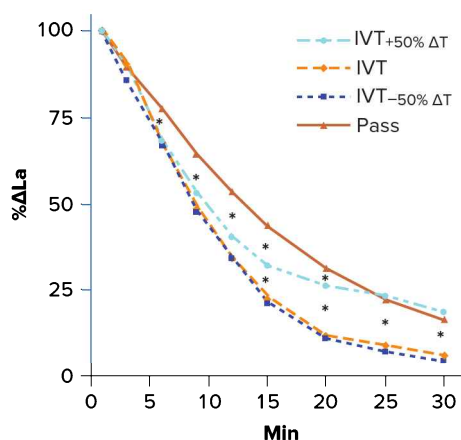


Figure 13.1 Blood lactate concentrations following exercise. Mean blood lactate (La) concentration during passive (Pass) and active recoveries at: (1) 50% of the difference (50%DT) between individual anaerobic threshold and individual ventilatory threshold (IVT) above the IVT (IVT+50%DT), (2) IVT recovery intensity and (3) 50%DT below the IVT (IVT50%DT). Comparison between recovery conditions within each time point for soccer players. *Significant difference versus IVT+50%DT recovery intensity (FROM BALDARI ET AL. 2004¹⁶)

greater perceived recovery. These findings highlight that players view recovery as an important part of their training and that AFL players tend not to select active recovery. The authors suggested that this may be due to players not wanting to expend further energy.

Psychological effects of active recovery

While metabolite elimination has been the primary focus of many studies, some authors have reported positive psychological effects of active recovery. Le Meur and Hausswirth have suggested that stimulation of metabosensitive type III and IV afferent nerves may facilitate better mental recovery.²³ In a study investigating the effect of active recovery following rugby matches, there were reductions in a measure of mental wellbeing, the POMS scores, in players who performed low-intensity water-based exercise when compared to passive recovery.²⁴ We cannot generalise these findings as the study measured only 15 players after one match.

In summary, active recovery most effectively reduces blood lactate levels at exercise intensity close to the lactate threshold; such reductions may improve performance for a subsequent exercise bout within 10–20 minutes. There is no strong evidence for the use of active recovery for exercise bouts that are several hours or days after the initial session. Note that active recovery has never been reported to impair performance.

MASSAGE

Massage has become ubiquitous in sports recovery. The effectiveness of massage has been attributed to increases in blood flow, mechanical effects on muscle tissue, reduced pain and psychological benefits.

Massage and blood flow

One of the postulated mechanisms by which massage may speed recovery is through increased local blood flow to remove metabolites and mobilise inflammatory markers. Studies that investigated the influence of massage on muscle blood flow and blood lactate removal reported no improvement in blood flow or blood lactate levels following either superficial or deep massage of the quadriceps after high-intensity cycling. They did, however, observe improved blood flow following active recovery.¹²

In the study by Wiltshire et al. of the effect of massage of the forearm after a fatiguing isometric grip protocol, massage *reduced* blood flow.²⁵ This may have been due to mechanical pressure on blood vessels. While no other studies have reported blood flow reductions with massage, many studies failed to detect any effect on either blood flow or lactate levels.^{26–28} There is strong evidence that, when compared to active recovery, massage is no more effective in reducing blood lactate than passive

control activity. As discussed earlier in this chapter, active recovery is well established as an effective way to facilitate blood flow. Therefore, if the goal of the recovery is to increase local blood flow, active recovery should be the modality of choice. Although some might argue that the lack of effect of massage on blood flow means massage does not assist recovery, massage may address other aspects of recovery.

Massage, muscle tone and viscoelasticity

Many recovery interventions, particularly the manual therapies, aim to influence the mechanical properties of soft tissues. Athletes and clinicians report reduced muscle tone and spasm following massage. The assumption is that this will reduce soreness, permit force production and enhance quality of movement. However, the influence of soft tissue mechanical properties on performance and recovery has not been well characterised. Massage may reduce tissue stiffness.^{29, 30} Although a number of studies have considered the effect of compression on muscle viscoelastic behaviour, there are limited studies investigating the effects of massage on these factors.³⁰ Thus, the effect of massage on these factors following exercise and the potential performance benefits are not clear.

Massage and delayed onset muscle soreness (DOMS)

Massage in post-exercise recovery is often assessed using a delayed onset muscle soreness (DOMS) model to determine effects on soreness and measures of muscle performance. While there are challenges extrapolating findings from these studies due to the diverse exercise regimens and massage protocols employed, a few important trends emerge. A number of studies report that massage is effective in reducing post-exercise soreness; however, its effectiveness in improving measures of muscle function is less positive. Smith et al. observed significant reductions in soreness of the elbow flexors following eccentric exercise in participants who received a 30-minute massage 2 hours after exercise; reductions in pain were apparent from 24 to 96 hours post-exercise.³¹ Similarly, Zainuddin et al. reported that massage applied for 10 minutes at 3 hours after exercise reduced pain, swelling and serum CK levels when compared to controls.³² No such improvements were observed for recovery of muscle performance (maximum voluntary contraction). A positive effect on soreness but not muscle function was also observed for 20 minutes of massage administered to the hamstrings 2 hours after eccentric exercise when compared to sham massage.³³

Positive effects for both muscle performance and soreness have been reported when massage is combined with other modalities. Jakeman, Byrne and Eston compared

a control group with a combination of compression garments and a 30-minute massage administered immediately after exercise.³⁴ Athletes who received massage and wore compression garments reported less pain as well as improved CMJ, squat jump and isokinetic muscle strength, compared to control athletes. A further study in basketball players using a post-match recovery strategy of combined stretches and massage reported reduced soreness and increased CMJ in male players, while female players demonstrated improvements in sprint performance only.³⁵

PRACTICE PEARL

The beneficial effects of massage on muscle recovery appear to be greatest when treatment is administered within two hours of exercise; there is no evidence to support the use of massage for recovery 24 hours after exercise has ceased.³⁶ This makes clinical sense: most recovery modalities discussed in this chapter should be administered as soon as possible after exercise.

Cellular and structural effects of massage

There is some evidence that massage may influence key cellular mediators. Rapaport et al. reported increases in circulating phenotypic lymphocyte markers and decreased mitogen-stimulated cytokine production following massage.³⁷ Similarly, Crane et al. observed a downregulation of pro-inflammatory cytokine expression following massage after eccentric exercise.³⁸ The authors also reported increased activity of mechanically sensitive proteins (FAK and ERK1/2). These observations suggest that the mechanical stimulus provided by massage may be capable of influencing mechanotransduction in the muscle (see Chapter 17). In a series of studies using a rabbit model, researchers from Ohio State University investigating the effects of massage-like compressive loading following eccentric exercise have demonstrated accelerated recovery of isometric torque production,³⁹ altered viscoelastic properties,³⁰ decreased number of torn muscle fibres and enhanced restoration of muscle function.⁴⁰ They have also identified a number of key variables that may be important to the use of massage in recovery. These have included timing,⁴¹ magnitude and frequency of the applied force, and treatment duration.³⁹

These findings provide a conceptual framework for some of the effects of massage; however, further studies are required to determine their clinical significance.

Psychological effects of massage

Consistent psychological benefits of massage are reported in the literature with positive effects reported for salivary IgA, POMS and perceived recovery of fatigue from exercise. Moraska observed that massage in non-athletic populations reduces psychological measures including anxiety, tension, stress and depression, while increasing mood and quality-of-life measures.⁴² Arroyo-Morales et al. suggest that massage restores autonomic nervous system balance via its effects on the parasympathetic nervous system.⁴³ They observed that a 40-minute massage applied after three 30-second Wingate tests restored or maintained pre-exercise heart rate variability values when compared to sham ultrasound treatment. Massage also facilitates relaxation and has been demonstrated to improve scores on the General Wellbeing Scale (an 18-item scale with subscales measuring anxiety, depression, positive wellbeing, self-control, vitality and general health) and the Perceived Stress Scale in healthy adults.⁴⁴

The type of massage used can influence the degree of relaxation. In one study comparing light, moderate and vibratory stimulation, moderate massage was demonstrated to offer greater reduction in stress with a concomitant decrease in heart rate and EEG.⁴⁵ Interestingly, the light and vibratory stimulation groups showed increased arousal as evidenced by increased heart rate and EEG activity. It is therefore important to consider the type of massage intervention used.

Massage is highly valued by athletes. Studies in both athletic and non-athletic populations have highlighted that one of the main reasons for using massage therapy is that it provides a personalised approach to care that enhances relaxation and creates a positive client-therapist relationship.⁴⁶ The 'culture' of a massage experience (how professional the setting, the relationship with the clinician) influences the athlete's perception of the massage. There is strong evidence that massage does not increase blood flow or aid metabolite removal, but it may influence various inflammatory and cellular responses. There is no evidence that massage is detrimental to recovery.

PRACTICE PEARL

Massage effectively modifies both perception of recovery and soreness. The effect of massage on correlates of performance is less clear and may be strongly influenced by the inclusion of other recovery strategies.

NEUROMUSCULAR ELECTRICAL STIMULATION

Neuromuscular electrical stimulation (NMES) is a modality commonly used to promote recovery in athletes. A review of NMES in post-exercise performance recovery highlighted the challenges associated with determining its effectiveness due to the wide variety of exercise types as well as stimulation protocols used.⁴⁷ In general, the application of electrical stimulation following exercise is proposed to have two primary effects:

1. increased blood flow and metabolite removal through muscle pump action
2. reduction in pain through stimulation of sensory nerve endings.

While attractive in its assumptions, the evidence regarding the ability of NMES to effectively increase local blood flow and reduce blood lactate levels is less convincing.

NMES and blood flow

The importance of the muscle pump to facilitate blood flow in the lower limb is well established. The calf muscle has been described as the second heart⁴⁸ and the effectiveness of low-level calf muscle activity to improve local blood flow has been identified.^{49, 50} A number of studies have compared the effect of NMES with active recovery and have consistently shown that active recovery is more effective in increasing blood flow and reducing blood lactate levels. In a study comparing the effects of four modalities (active recovery, passive recovery, NMES and CWI) on the recovery of grip strength following a fatiguing rock-climbing protocol, Heyman et al. reported no difference in blood lactate levels between NMES and passive recovery while observing a significant increase in blood flow and a concomitant reduction in blood lactate following 20 minutes of gentle cycling.¹⁸

Some positive effects on lower limb blood flow for NMES have recently been reported using devices that focus on peroneal nerve stimulation. These studies derive from the use of electrical stimulation in angiology and highlight the importance of deep calf muscle activity. In two studies investigating peroneal nerve stimulation following a 90-minute shuttle test and a Yo-Yo intermittent recovery test,^{51, 52} increases in local blood flow were reported. These observations suggest that NMES of this nature may enhance metabolite clearance following exercise when compared to passive recovery. However, it is not clear how these interventions compare with an active recovery protocol and whether they correspond to improvements in performance.

NMES and performance

The effect of NMES on correlates of sports performance such as strength, endurance and jump performance has been assessed following various types of exercise. While isolated studies have indicated that NMES may be slightly more effective than passive recovery for restoration of strength⁵³ and running performance,⁵² the majority of authors conclude that NMES is no more effective than passive recovery for these parameters. Although the lack of consistent positive effects of NMES may be due to the wide range of methods of stimulation used following different exercise protocols, it is not possible to state whether NMES effectively restores performance following exercise.⁴⁷

NMES and muscle soreness

As outlined above, reducing muscle soreness is a common aspect of sports recovery. A number of studies have investigated the effect of NMES on DOMS. Most studies have failed to observe any positive effect of NMES on soreness when compared to active or passive recovery.⁴⁷ A few studies have suggested that NMES may temporarily reduce pain,^{54, 55} but these effects do not correlate with improved performance. Maffiuletti and DuPont make an interesting observation that NMES can effectively improve the subjective perception of recovery in athletes.⁵⁶ The authors highlight that athletes consistently report that they feel better recovered or have greater levels of perceived energy than those who do not use NMES despite displaying no associated improvements in sports performance. The role of altered pain perception and other psychological factors in this regard is unclear; however, as discussed below, the perception of recovery may have beneficial effects for performance.

In summary, there is some limited evidence that NMES can enhance blood flow and metabolite removal with certain types of electrical current applied over the peroneal nerve. There is no evidence that NMES can effectively enhance recovery of correlates of performance, but it may effectively enhance the perception of recovery through neuropsychological means. A pragmatic approach to the use of NMES in recovery is shown in Table 13.1.

STRETCHING

Athletes often stretch after exercise in an attempt to improve range of motion and reduce the perception of musculotendinous stiffness. While it is a regular component of post-exercise regimens, there is limited evidence of the effect of stretching on various aspects of recovery.

Table 13.1 A pragmatic approach to the use of neuromuscular electrical stimulation (NMES) in recovery⁵⁶

- NMES should be combined with submaximal voluntary dynamic contractions (such as toe curls) whenever possible, in order to maximise blood flow increase.
- NMES should be applied distally (to the calf or foot muscles and eventually to the common peroneal nerve) rather than proximally (to the quadriceps muscle belly) to maximise the muscle pump effect.
- NMES is most effective in weight-bearing and contact sports to reduce signs and symptoms of muscle damage.
- NMES should be delivered with low doses but for long durations (such as overnight), during travel or by means of electrostatically charged self-adhesive membranes.

Maffiuletti N, Dupont G. Electrostimulation-related recovery strategies. *Aspetar Sports Medicine Journal* 2015;4:64–70. Reproduced with permission.

Lund et al. suggested that stretching following bouts of eccentric exercise may delay recovery.⁵⁷ In a study investigating the effect of static stretching on DOMS following eccentric exercise in the quadriceps of seven untrained females, they reported that recovery of strength was impaired in the group who stretched their quadriceps for three repetitions of 30 seconds each day after exercise. The authors proposed that stretching after eccentric exercise caused further mechanical disruption and exacerbated muscle damage. In contrast, Torres et al. reported no effect of daily stretching on maximum voluntary contraction of the quadriceps following eccentric exercise in healthy untrained men.⁵⁸ They also reported no effect of static stretching on CK levels or soreness, but did observe improvements in muscle stiffness when compared to passive recovery. The authors suggest that the changes in muscle stiffness may have functional benefits in terms of normalising muscle tone and enhancing quality of movement. This is an interesting postulation and merits further research.

A few studies have reported positive effects of stretching on cycling performance; Dawson et al. noted improvements in a 6-second cycle sprint test 15 hours after an Australian rules football match with stretching when compared to a control.⁵⁹ Dynamic stretching also improved performance during a second exhaustive bout of cycling.¹¹ Given the isolated nature of these findings and the fact that they were only observed during cycling, it is not possible to state that stretching positively influences recovery of performance in other sport settings.

Most studies report no effect of stretching on recovery. A comparison of static stretching, CWI and passive recovery following strenuous stair climbing in rowers reported no differences between each of the interventions for strength or CK levels.⁶⁰ Similarly, no differences were observed for systemic markers of stress and inflammation, blood lactate, anaerobic performance or psychological mood states after an exhaustive bout of endurance exercise for static stretching or for active recovery using either an anti-gravity treadmill or cycle ergometer.⁶¹ Reviews have consistently failed to detect any positive effect of stretching on correlates of recovery after exercise despite testing various different stretching protocols.⁶² It

is important to note that post-exercise stretching does not appear to have any detrimental effects on performance.

SLEEP

Optimal sleep quality and quantity is the single best recovery strategy available to athletes. In studies of performance following partial sleep deprivation, sustained exercise was more affected than single maximal efforts.⁶³ Thus, longer submaximal exercise tasks such as long-distance running may be affected following sleep deprivation.

Suboptimal sleep may have a range of consequences for athletes beyond reduced performance, such as impaired immunity and endocrine function. This can affect the recovery process and adaptations to training.⁶⁴ Sleep deprivation can also result in reduced cognitive function, increased pain perception, mood changes, altered metabolism and changes in inflammatory indicators.⁶⁴

Athletes often rank sleep deprivation and poor sleep quality as prominent causes of fatigue and tiredness.⁶⁵ Many athletes often have difficulties sleeping prior to important competitions.^{66, 67} Juliff et al. questioned 283 elite Australian athletes and found that 64.0% of athletes indicated worse sleep on at least one occasion in the nights prior to an important competition over the past 12 months.⁶⁷ Athletes complained of difficulty falling asleep (82.1%) particularly because of thoughts about the competition (83.5%) and general nervousness (43.8%).

Activity monitors have recently been utilised to measure sleep in British⁶⁸ and Australian⁶⁹ athletes with quite similar results. The 47 elite British athletes spent a total of $8:40 \pm 0:50$ hours:minutes in bed, compared with $8:10 \pm 0:20$ in the control group. The athlete group had a longer sleep latency (time taken to fall asleep) (18 ± 17 vs. 5 ± 3 min) and a lower sleep efficiency (estimate of sleep quality) than controls (81 ± 6 vs $89 \pm 4\%$), resulting in a similar time asleep ($6:55 \pm 0:43$ vs $7:11 \pm 0:25$ hours:minutes; mean $\pm$ SD). Thus, athletes had a comparable quantity of sleep to controls, but inferior quality of sleep.

On average, the 124 Australian individual and team sport athletes went to bed at $23:59 \pm 1.3$, woke up at $07:15 \pm 1.2$ and obtained 6.8 ± 1.1 hours of sleep per night. Athletes from individual sports went to bed earlier,

Table 13.2 Sleep hygiene strategies. Adapted from Halson⁶⁴

Bedroom	• The bedroom should be cool (19–21°C is best), dark and quiet. • A comfortable bed and pillows is important.
Electronics	• Avoid watching television in bed and using the computer or smartphone in bed. These can steal sleep time and form bad habits.
Food and fluid	• Avoid the use of caffeinated food and fluids later in the day. • Do not go to bed after consuming too much fluid; this may result in waking up to use the bathroom.
Behaviour	• Create a good sleep routine by going to bed at the same time and waking up at the same time. • Before bed routine can help the body prepare for sleep. The routine should start about 30 minutes before bedtime (i.e. clean teeth, read a book, etc.). • Avoid watching the clock. Many people who struggle with sleep tend to watch the clock too much. Frequently checking the clock during the night can wake you up (especially if you turn on the light to read the time) and may reinforce negative thoughts. • Be organised. Utilise a 'to-do' list or diary to ensure organisation and avoid unnecessary over-thinking while trying to sleep. • Investigate relaxation/breathing techniques. • If you haven't been able to get to sleep after about 20 minutes or more, get up and do something calming or boring until you feel sleepy, then return to bed and try again. Sit quietly on the couch with the lights off (bright light will tell your brain that it's time to wake up).

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woke up earlier and obtained less sleep (individual vs team; 6.5 vs 7.0 h) than athletes from team sports.⁶⁹

Given the effects of sleep deprivation on performance and other important aspects of elite athlete physiology and psychology, athletes should be appropriately educated regarding the importance of sleep and potential strategies to enhance sleep. Sleep hygiene practices that may enhance sleep quantity and quality are outlined in Table 13.2.

WATER IMMERSION

Post-exercise water immersion of various types has become popular with elite athletes to accelerate recovery from training and competition. Three main forms of water immersion exist: cold water immersion (CWI), hot water immersion (HWI) and contrast water therapy (CWT). Several authors have conducted meta-analyses to help clarify and explore the effects of these modalities on performance. Poppendieck et al. examined effect sizes for a range of performance measures following various cooling strategies.⁷⁰ The largest average effect size was for sprint performance (2.6%, $g=0.69$), followed by endurance parameters (2.6%, $g=0.19$), jump performance (3.0%, $g=0.15$) and strength (1.8%, $g=0.10$). Similarly, Bieuzen et al. examined the effects of CWT on exercise-induced muscle damage and reported significantly greater improvements in CWT when compared to passive recovery.⁷¹ Further, CWT reduced muscle strength loss compared to passive recovery. Studies that compare the different forms of water immersion typically report that CWI and CWT enhance recovery more than HWI when used appropriately following cycling^{72, 73} and

resistance training.⁷⁴ Studies report accelerated performance recovery following CWI and CWT compared to HWI.⁷⁵

PRACTICE PEARL

The results of recent meta-analyses suggest that on average CWI and CWT recovery interventions are beneficial for recovery of performance and perceptual parameters.

As a result, recent research has focused on possible underlying mechanisms to explain these improvements.

Cold water immersion and other forms of cryotherapy have traditionally been utilised to minimise inflammatory responses, and decrease oedema and pain. Physiologically, this has been shown to occur through reductions in limb blood flow, and reductions in skin, muscle and core temperature.⁷⁶ Further, hydrostatic pressure can limit swelling and fluid accumulation, increase central blood volume and facilitate removal of metabolic waste.⁷⁷ Positive cardiovascular effects have also been observed with CWI, including lower heart rate, increased cardiac output and/or stroke volume.⁷⁷

The effects of CWI on both the brain and pacing strategies have also been examined as possible contributors to improved performance findings. De Pauw et al. used brain mapping via electroencephalography to demonstrate changes in brain electrical activity indicative of decreased arousal and increased somatosensory information processing.⁷⁸ The same research group also

demonstrated different cycling time trial pacing strategies in the heat,⁷⁹ likely the result of decreased thermal strain.

For recovery techniques to be beneficial following training, they must not only accelerate recovery, but also allow adaptations to training to occur. Based on the improvements in performance observed in acute CWI studies, two adaptation theories have been proposed. The first is that hydrotherapy allows athletes to perform subsequent training sessions with a greater training load or quality, thus resulting in an enhanced stimulus for adaptation. Conversely, the second theory suggests that CWI may decrease adaptations to training due to minimisation of fatigue and inflammation occurring following training. Thus, some practitioners are questioning the use of chronic hydrotherapy treatment in elite athletes with respect to a potential negative influence on training adaptation.

Yamane et al. report that regular CWI resulted in an attenuation of cycling performance improvements when compared to a control condition over a training period of 4–6 weeks.⁸⁰ Similar findings were reported by Higgins et al. whereby repeat sprint test performance decreased with CWI when compared with control in amateur rugby union players.⁸¹ Further, Frohlich et al. reported a small reduction in single-leg adaptation to strength training in trained male students when compared to cases where the leg was not immersed.⁸²

In contrast, recent research using elite cyclists found that CWI had no effect on adaptation during 6 weeks of training.⁸³ CWI has also been shown to better maintain cycling performance over 5 days.^{84, 85} When examining submaximal performance, two recent studies have demonstrated an increase in submaximal cycling⁸⁶ and submaximal resistance exercise⁸⁷ suggesting that an athlete may be able to train at higher workloads, thereby increasing long-term training adaptation. This is an important new line of investigation as previous studies have focused almost entirely on maximal performance and not on the ability to enhance daily training.

Increasing evidence exists to support the use of CWI and CWT, particularly in acute settings such as that of a competition or event. Further research is needed to fully elucidate the longer-term effects of water immersion on adaptation to training.

COMPRESSION

Many recovery strategies for elite athletes are based on medical equipment or therapies originally used on patients. Compression clothing is one of these strategies as it has traditionally been used to treat various lymphatic and circulatory conditions.⁸⁸ Compression garments are thought to improve return of blood to the heart through application of graduated compression to the limbs from the ankle upwards. The external pressure created by

compression garments may reduce swelling, inflammation and muscle soreness, and thereby enhance recovery.⁸⁹

Research investigating compression in athletes typically focuses on wearing compression either during exercise or during recovery. Research examining performance while wearing compression is conflicting, mainly due to the heterogeneity in subjects, study design and type of compression worn. In a recent meta-analysis, Born et al. reported small effect sizes for the application of compression clothing during exercise for short-duration sprints (10–60 m), vertical-jump height, extending time to exhaustion (such as running at $\dot{V}O_{2\max}$ or during incremental tests) and time trial performance (3–60 min).⁹⁰

From a specific recovery perspective, compression has been shown to improve performance between two 40 km time trials performed 24 hours apart.⁹¹ In this case, the compression was worn for the full 24 hours, which may not be practical or comfortable for many athletes. Compression has also been shown to improve sprint performance, with a shorter recovery period. Maximal sprint performance was enhanced when using compression for 20 minutes between sprints.⁹² Results of a meta-analysis suggest that when compression clothing was applied for recovery purposes after exercise, small to moderate effect sizes were observed in recovery of maximal strength and power, especially vertical-jump exercise; reductions in muscle swelling and perceived muscle pain; blood lactate removal; and increases in body temperature.⁸⁸ Research investigating a multifaceted approach to recovery (including the use of full-body compression garments, CWI and sleep) was shown to assist subsequent tennis performance as well as resulting in favourable physiological outcomes such as reduced blood lactate, heart rate and perceived muscle and joint soreness.⁹³

From the research available, it appears that compression garments may be beneficial and do not appear to be harmful in any way. We recommend that compression garments be worn for at least 60 minutes after training or competition to maximise the recovery process.

NUTRITION

with GRAEME CLOSE

Nutrition is essential for complete recovery. Recovery eating and drinking should help meet the athlete's total nutritional goals in terms of energy needs, desire to manipulate physique and overall requirements for nutrients. It must also be manageable within the practical constraints that govern the athlete's food intake; for example, to be affordable, to fit in with the daily timetable and social commitments, and to involve foods and drinks that are acceptable and readily available. Three primary concerns are fluid replacement, restoring fuels (largely carbohydrate in competitive sport today) and repair (largely protein).

Replacing fluids

Substantial fluid loss should be replaced. The simplest way to determine the amount of fluid lost is by weighing the athlete pre- and post-event. Every 1 kilogram of weight loss equates to 1 litre of fluid loss. The athlete should consume (i.e. via drinks and meals) water and sodium at a modest rate that minimises diuresis/urinary losses.⁹⁴ Because sweat losses and obligatory urine losses continue during the post-exercise phase, effective rehydration requires the intake of about 125–150% of the final fluid deficit (e.g. 1.25–1.5 L fluid for every 1 kg of body weight lost).⁹⁵

Dietary sodium/sodium chloride (from foods or fluids) helps to retain ingested fluids, especially extracellular fluids, including plasma volume. Therefore, athletes should include sodium in their post-exercise nutrition, particularly when large sodium losses have been incurred. The optimal sodium level in a rehydration drink appears to be close to the sodium content of sweat⁹⁶ and thus may be as high as 50–80 mmol/L; this is higher than the sodium concentration of typical sports drinks and is close to the level found in oral rehydration solutions manufactured for the treatment of diarrhoea. Fluid can be consumed with meals or snacks that include salt-rich foods (e.g. breads, breakfast cereals, crackers, cheese, preserved meats) or have been seasoned with added salt. Milk is a better sport drink for rehydration than many commercial products because of its sodium content as well as the protein and fat in it which may aid fluid retention.⁹⁷

Alcohol, particularly strong beers and shots, is a diuretic and is thus discouraged in the recovery period. However, the previous warnings about caffeine as a diuretic appear to be overstated when it is habitually consumed in moderate amounts (e.g. <180 mg = 1–2 standard brewed cups).⁹⁵

The athlete should start to consume fluids on finishing and aim to consume the target volume over the next 2–4 hours. Spreading fluid intake over a period of time promotes gastrointestinal comfort and maximises retention via smaller urine losses.⁹⁸ When the athlete's total energy needs are low, low-energy fluids such as water, low-energy soft drinks and mineral waters are recommended. When energy needs are higher, fluids can be combined with energy sources.

Replacing fuel

As most athletes use carbohydrates as their major fuel supply, glycogen restoration is one of the goals of post-exercise recovery, particularly between bouts of carbohydrate-dependent exercise where there is a priority on performance in the second session. Refuelling requires adequate carbohydrate intake and time. Provided that total energy intake is adequate,⁹⁹ increasing carbohydrate intake increases muscle glycogen storage until an upper limit for glycogen synthesis is reached.¹⁰⁰ The most recent guidelines for post-exercise glycogen storage recognise a scaling of requirements according to the fuel cost of training or competition and the athlete's body size.¹⁰¹ (Figure 13.2)

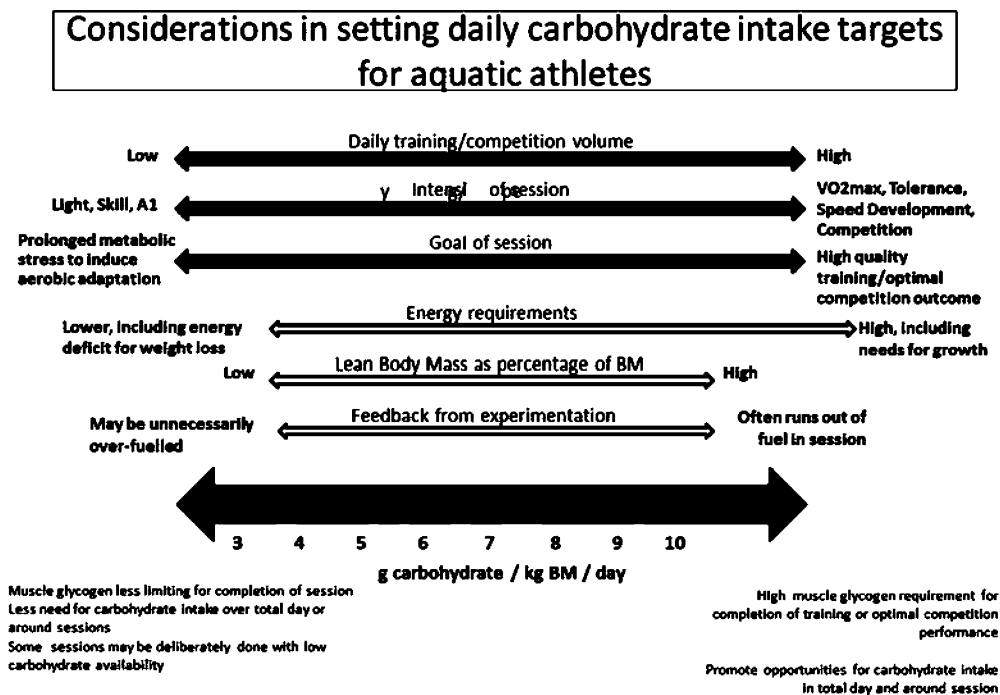


Figure 13.2 Considerations in setting daily carbohydrate intake targets for aquatic athletes⁹⁸

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Table 13.3 Carbohydrate-rich recovery snacks (50 g CHO portions)¹⁰⁴

- 700–800 mL sports drink
- 2 sports gels
- 500 mL fruit juice or soft drink
- 300 mL carbohydrate loader drink
- 2 slices toast/bread with jam or honey or banana topping
- 2 cereal bars
- 1 cup thick vegetable soup + large bread roll
- 115 g (1 large or 2 small) cake-style muffins, fruit buns or scones
- 250 g baked potato with salsa filling
- 100 g pancakes (2 stack) + 30 g syrup

The mean hourly rate of glycogen restoration is about 2% of the total glycogen capacity per hour; the maximum rate is around 5%. Thus, it can take approximately 24 hours to normalise stores after substantial levels of depletion.⁹⁸ During the 2–4 hour period after exercise, there is a potential for high rates of muscle glycogen storage as a result of the depletion-activated stimulation of the glycogen synthase enzyme and exercise-induced increases in muscle membrane permeability and insulin sensitivity.¹⁰² This potential can be realised only if carbohydrates are consumed during this period; if not, refuelling rates are very low.¹⁰³

Meals and snacks can be chosen from a variety of foods and fluids according to personal preferences of type and timing of intake.^{100, 101} The athlete should aim for recovery drinks, snacks or meals providing carbohydrate equal to approximately 1 g/kg/hour body mass for the first 4 hours post-exercise. Appropriate carbohydrate-rich foods are shown in Table 13.3.

Repair

The Australian Institute of Sport (AIS) provides the following information in its factsheet on recovery nutrition:

*Prolonged and high-intensity exercise causes a substantial breakdown of muscle protein. During the recovery phase there is a reduction in catabolic (breakdown) processes and a gradual increase in anabolic (building) processes, which continues for at least 24 hours after exercise. Early intake after exercise (within the first hour) of essential amino acids from good quality protein foods helps to promote the increase in protein rebuilding. Consuming food sources of protein in meals and snacks after this 'window of opportunity' will further promote protein synthesis, though the rate at which it occurs is less.*¹⁰⁴

The maximal protein synthetic response to a resistance exercise bout is achieved with the intake of approximately 20–25 g of high-quality protein¹⁰⁵ or 0.3 g/kg body mass.

Protein consumed in excess of this threshold stimulates increased rates of irreversible protein oxidation and may be considered wasteful.¹⁰⁵ The pattern of intake over the rest of the day should probably involve a series of meals or snacks every 3–5 hours providing about 20–25 g of protein rather than frequent feedings of smaller protein doses or infrequent meals with larger protein doses.¹⁰⁶ Consuming protein just before bed may also be valuable because it can increase muscle protein synthesis at a time when it would otherwise be low.¹⁰⁷

The important characteristics of protein-rich foods appear to be the digestibility of the food and its content of essential amino acids, especially leucine. High-quality protein-rich foods such as milk (or whey), eggs and meat are all associated with significant increases in muscle protein synthesis after exercise.¹⁰⁸ In the period immediately after exercise, at least, there is superior muscle protein synthesis with the intake of proteins that are rapidly digested to provide a rapid rise in plasma leucine concentrations; this includes whey protein (a rapidly digested, or 'fast', protein) and liquid protein forms.¹⁰⁹

Adding a source of carbohydrate to the post-exercise protein will further enhance the training adaptation by reducing the degree of muscle protein breakdown. Chocolate milk is an excellent recovery drink combining carbohydrate and protein.¹¹⁰ Table 13.4 provides a list of carbohydrate rich snacks that also provide at least 10 g of protein, while Table 13.5 lists a number of everyday foods that provide approximately 10 g of protein.¹⁰⁴

According to the Australian Institute of Sport (AIS):

*Some athletes finish sessions with a good appetite, so most foods are appealing to eat. On the other hand, a fatigued athlete may only feel like eating something that is compact and easy to chew. When snacks need to be kept or eaten at the training venue itself, foods and drinks that require minimal storage and preparation are useful. At other times, valuable features of recovery foods include being portable and able to travel interstate or overseas. Situations and challenges in sport change from day to day and between athletes, so recovery snacks need to be carefully chosen to meet these needs.*¹⁰⁴

Increasing numbers of athletes, especially endurance and ultra-endurance, are using fat rather than carbohydrate as their preferred primary fuel source.¹¹¹ Others are adopting the 'train low, compete high' regime which involves low carbohydrate intake during training with the aim of optimising use of fat as a fuel source, then using carbohydrates on the day of competition.^{112, 113}

In those athletes using the low-carbohydrate high-fat method of fuelling, glycogen stores are not significantly depleted so restoration with a high-carbohydrate meal following exercise is not required. Most athletes on this

Table 13.4 Carbohydrate-protein recovery snacks (contain 50 g CHO plus valuable source of protein and micronutrients)¹⁰⁴

- 250–300 mL liquid meal supplement
- 300 g creamed rice
- 250–300 mL milkshake or fruit smoothie
- 600 mL low-fat flavoured milk
- 1–2 sports bars (check labels for carbohydrate and protein content)
- 1 large bowl (2 cups) breakfast cereal with milk
- 1 large or 2 small cereal bars + 200 g fruit-flavoured yoghurt
- 220 g baked beans on 2 slices of toast
- 1 bread roll with cheese/meat filling + large banana
- 300 g (bowl) fruit salad with 200 g fruit-flavoured yoghurt
- 2 crumpets with thick spread peanut butter + 250 mL glass of milk
- 250 g baked potato + cottage cheese filling + glass of milk

Table 13.5 Foods providing approximately 10 g of protein¹⁰⁴

- Animal foods
 - 40 g of cooked lean beef/pork/lamb
 - 40 g skinless cooked chicken
 - 50 g of canned tuna/salmon or cooked fish
 - 300 mL of milk/glass of Milo
 - 200 g tub of yoghurt
 - 300 mL flavoured milk
 - 1.5 slices (30 g) of cheese
 - 2 eggs
- Plant-based foods
 - 120 g of tofu
 - 4 slices of bread
 - 200 g of baked beans
 - 60 g of nuts
 - 2 cups of pasta/3 cups of rice
 - ¾ cup cooked lentils/kidney beans

regime find that simply having their normal meals is sufficient for nutritional recovery.

Before leaving the topic of fuel, we remind you that research funding influences which elements of sports medicine are addressed. Food companies have openly contributed to the advancement of science with research grant funding to quality scientists. Such funding helps to leverage national health research grants; for example, the National Institutes of Health (NIH) in the United States, the Medical Research Council (MRC) in the UK and the National Health and Medical Research Council (NHMRC) in Australia. Many of these grants were used to study the effects of carbohydrate on recovery and performance. Other industries, such as the dairy industry and the meat

industry, contributed to scientific research but their funding has not matched that of companies whose profits derive largely from carbohydrate products.

PULLING THE DIFFERENT THREADS TOGETHER—PRACTICAL CONSIDERATIONS FOR THE CLINICIAN

As stated above, recovery modalities for athletes must not only accelerate recovery but also allow training adaptation. Different types of activities place different demands upon the system and as such may require distinct approaches. A sound understanding of the nature of the activity from which the athlete seeks to recover and the associated metabolic, neural, psychological and mechanical load is necessary to ensure that each aspect of recovery is maximised. Table 13.6 summarises some of the factors that influence decision making.

For example, reduced-impact or non weight-bearing exercise may be preferable following high-intensity exercise that involves lower limb mechanical loading. It is reasonable to consider that the tissue damage incurred during sports such as rugby may repair best with reduced mechanical loading; therefore, such strategies including cycling or water-based recovery may be more appropriate.

The perception of recovery appears to be important and has the potential to influence sports performance. Investigating the effect of the perception of soreness on vertical jump in male soccer players, Smith and Jackson reported an inverse relationship between perception of soreness and vertical jump performance.¹¹⁴ This poses the interesting suggestion that performance during many of the tasks associated with recovery literature may be positively influenced by perception which, one would postulate, is related to central neuropsychological mechanisms rather than to local physiological effects. Given that many of the modalities commonly used as part of recovery have the potential to provide a strong psychological response, the positive effects reported by authors may include a large central effect. This should not be underestimated and specific strategies that maximise the complex interaction between perception, arousal and sports performance should be considered.

This is not without its challenges, as the ability to extrapolate these principles is limited and highly context-specific. For example, the increased stress associated with important matches or during a period of particularly high training load may have significant effects on the perception of fatigue and subsequent performance. It is reasonable to suggest that during periods such as these, various modalities that facilitate enhanced feelings of wellbeing and reduced pain may be preferable; such modalities may include massage, NMES and compression garments.

Table 13.6 Summary of the various recovery modalities

Modality	Systems influenced	Key recommendations
Active recovery	• Metabolic • Psychological	• Most effective when subsequent exercise bout is within 20 min • Exercise intensity close to lactate threshold
Massage	• Neural • Structural • Psychological	• Within 2 hours after exercise • Minimum 10 min duration • Moderate intensity
Neuromuscular electrical stimulation	• Neural • Psychological	• To increase blood flow, use distal application to initiate calf muscle pump • As soon as possible after exercise
Stretching	• Structural (muscle tone) • Psychological	• Static stretching immediately after exercise • Mild–moderate intensity
Cold-water immersion	• Psychological • Neural • Metabolic	• As soon as possible after exercise • A total duration of 10–15 min • Temperature range 10–15°C
Contrast water therapy	• Psychological • Neural • Metabolic	• As soon as possible after exercise • Cold: 10–15°C • Hot: 38–40°C • 5–7 rotations of 1–2 min
Compression	• Psychological • Neural	• Apply immediately after exercise for minimum of 60 min
Sleep	• All systems	• Use strategies outlined in Table 13.2 to enhance sleep quality
Nutrition	• Metabolic • Structural • Psychological	• Rehydrate to 1.5x the weighed fluid loss (to account for ongoing losses) • Refuel with appropriate carbohydrate (Tables 13.3 and 13.4) • Repair with protein (Tables 13.4 and 13.5)

Key considerations to aid decision making include the period of the annual training program (e.g. pre-season vs mid-season, high volume vs low volume), as well as the previous experience of the athlete and the culture of the specific sport. For example, CWI has become part of normal practice of many team sports and is now a usual part of the post-match recovery routine. Many such practices become embedded in preparation and scheduling; a recovery strategy that involves an alteration in the use of CWI has the potential to influence the perception of recovery as well as any associated psychological or physiological effects. Similarly, a decision to include specific interventions that are unfamiliar may place additional stress on the athlete. As such any changes (whether introduction or removal

of specific practices) should be carried out at a time that minimises any secondary effects on the athlete.

SUMMARY

Sleep and good nutritional practices are the cornerstone of recovery and every effort should be made to maximise the benefits of these interventions. As outlined above, additional modalities may be used to augment recovery through the complex interaction of psychological, metabolic, structural and neurophysiological factors. The ability to influence recovery in the sporting environment relies on the intelligent application of these principles to meet the specific needs of the athlete.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e